

CSE4204 — Mobile Computing Lab

Week 04 Submission

System Design & Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

Team Name	CSE4204-8D-T03
Section	8D
Project Title	Smart Farmer Assistant: AI-Based Agriculture Support System
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GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant
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Activity Diagram

The Activity Diagram illustrates the end-to-end workflow of the Crop Disease Detection feature — the system's most critical function. It uses three swimlanes to separate the Farmer's actions, the Django Backend's processing, and the AI Services layer. All decision points and alternate flows are modelled.

i. Disease Detection Workflow — Step-by-Step

Step	Swimlane	Action / Decision	Next
1	Farmer	Open application and navigate to Login page	Step 2
2	Farmer	Enter email and password → click Login	Step 3
3	Backend	DECISION: credentials valid? No → show error → back to Step 2	Step 4 (Yes)
4	Farmer	Enter Dashboard → select Disease Detection service	Step 5
5	Farmer	Upload crop leaf image (JPEG/PNG, max 5 MB)	Step 6
6	Backend	DECISION: image format/size valid? No → error → Step 5	Step 7 (Yes)
7	Backend	Pre-process image: resize to 224×224 px, normalise pixel values	Step 8
8	AI Services	TensorFlow CNN model runs inference on pre-processed image	Step 9
9	AI Services	DECISION: confidence $\geq 75\%$? No → return 'Unidentified'	Step 10 (Yes)
10	AI Services	Return disease name + confidence score to Backend	Step 11
11	AI Services	Gemini AI generates treatment advice and fertilizer suggestion	Step 12
12	Backend	Save record to disease_history and fertilizer_recommendations	Step 13
13	Farmer	View full Disease Report (disease, confidence, treatment)	Step 14
14	Farmer	DECISION: open AI Chatbot for follow-up? No → End	Step 15 (Yes)

Step	Swimlane	Action / Decision	Next
15	Farmer	Ask follow-up question in Chatbot	Step 16
16	AI Services	Gemini AI generates contextual farming answer	Step 17
17	Farmer	Read answer → End	—

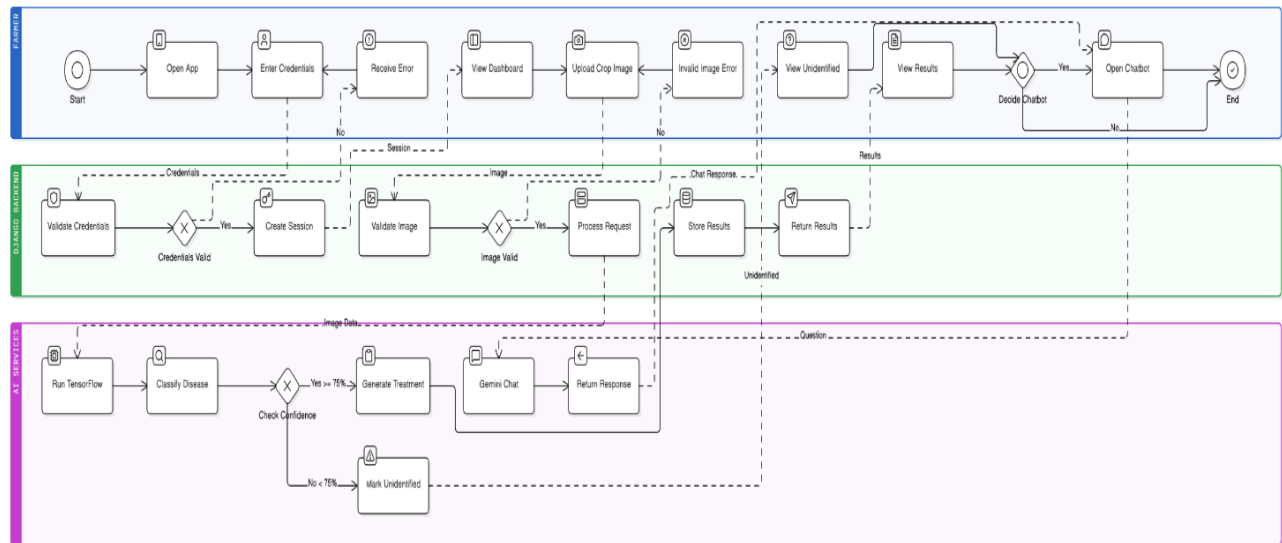


Figure 4: Activity Diagram — Crop Disease Detection Workflow (3-lane swim lane)